

MATH5320 – Homework X

1. Capital

For each type of position at a bank, tell whether it's an asset or a liability.

- (a) Cash
 - (b) Savings accounts
 - (c) Long bond positions
 - (d) Loans from another bank
 - (e) Mortgages made by the bank
 - (f) Long position in mortgage backed securities
 - (g) Office leases
 - (h) Payroll
-

Question 1:

This answer primarily refers to the slide below and is predicated from the accounting equation of $A = L + E$.

Capital

The basis of quantitative regulatory risk management is computing regulatory capital – how much capital a firm must maintain to be deemed “healthy”.

What is capital?

- Giving money out is buying something. Taking it in is selling something.
- Assets – Rights the firm bought – Everything the firm owns:
 - Purchased bonds, stocks, buildings, ...
- Liabilities – Rights the firm sold – Everything the firm owes:
 - Payroll, deposits, leases, ...

Position	Classification	Why
Cash	Asset	Banks owns the cash.
Savings Accounts	Liability	Despositor funds the bank owes
Long Bond	Asset	Bond is an asset.
Loans from another bank	Liability	Money borrowed by the bank from another bank, and is pending repayment.
Mortgages made by the bank	Asset	Loans owed to bank by homeowners
Long Positions in MBS	Asset	MBS are collateralised loans, that are bundled into a

		security that the bank can own.
Office Leases	Liability (Asset if prepaid)	Owes lease payments – technically an expense , but can be an asset <u>if prepaid.</u>
Payroll	Liability	Owes wages + salaries to employees

Question 2:

2. Risk weighted assets

A bank has the following assets and liabilities:

- 69000 in cash
- 73000 in residential mortgages
- 47000 in corporate bonds
- 182000 in deposits

Given risk weights of:

- 0% – Home country sovereign debt
- 20% – Certain public debt
- 35% – Interbank loans
- 45% – Residential mortgages
- 100% – Private sector debt

- (a) What are the assets and liabilities?
 - (b) What is the capital?
 - (c) What are the risk weighted assets?
 - (d) What is the capital ratio?
 - (e) What is the leverage ratio?
-

a)

For the given items, we know that the:

Assets in this case are given as:

- Cash – 69,000
- Residential Mortgages – 73,000
- Corporate bonds – 47,000

As an aggregate these are:

$$69,000 + 73,000 + 47,000 = 189,000$$

As a result, we have 182,000 in liabilities (from the deposits), and 189,000 in assets.

b)

Given the above, of 182,000 in liabilities and 189,000 in assets, we have 7,000 in Capital, or strictly speaking, Owner's Equity. Using the Accounting equation:

$$A = L + E$$

Therefore, from this equation, we have:

$$189,000 = 182,000 + E$$

$$\therefore E = 7,000$$

c)

From the above associated weights and assets:

- 69000 in cash
- 73000 in residential mortgages
- 47000 in corporate bonds
- 182000 in deposits

Given risk weights of:

- 0% – Home country sovereign debt
- 20% – Certain public debt
- 35% – Interbank loans
- 45% – Residential mortgages
- 100% – Private sector debt

We know that the assets are effectively a risk-weighting, as follows:

$$\Sigma A_i R_i$$

And then we have the following:

$$RWA = Cash * R_{Cash} + RM * R_{RM} + CB * R_{CB}$$

$$RWA = (73,000)(0.45) + (47,000)(1.0) = 79,850$$

Resultingly, we have an RWA of 79,850.

d)

We know that capital ratios are generally quoted through capital ratios, which are derived from the Accounting equation, as shown below:

Capital ratios	Risk weighted assets
Capital requirements are often quoted through capital ratios. A Assets L Liabilities E Equity (or capital) Basic balance sheet: $A - L = E$ Suppose we expect a default rate of p and a recovery rate of R. Expected loss rate $= I = p \times (1 - R)$. To stay solvent in the face of this expected loss, need: $\begin{aligned} \text{Assets} - \text{Losses} &> \text{Liabilities} \\ A - I \times A &> L \\ A - L &> I \times A \\ E &> I \times A \\ E/A &> I \end{aligned}$ To remain solvent in the face of such losses, need $E/A > I$.	Up to Basel 3, banks not using internal models had capital ratio requirements based on risk weighted assets. Example: - Cash position C, cash risk weight = 0%. - Corporate bond position B_c, risk weight = 100%. - Interbank loan position B_i, risk weight = 20%. Total risk weighted assets: $RWA = 0 \times C + 1 \times B_c + .2 \times B_i \quad (1)$ Minimum capital adequacy ratio: $E/RWA \geq 0.08 \quad (2)$ Then: $\begin{aligned} E &\geq .08(0 \times C + 1 \times B_c + .2 \times B_i) \\ &= .08 \times B_c + .016 \times B_i \end{aligned}$ Equivalent to requiring solvency in the face of 8% losses on B_c and 1.6% losses on B_i.

Thereby, in this case, if we consider the RHS case considering it with:

$$\frac{E}{RWA} = \frac{7,000}{79,850} = 0.0876$$

As a result, the given capital ratio, will be about 8.77%. As such, under the lecture's benchmark, this is above the 8% Total Capital Ratio under Basel II and the US.

Capital and liquidity ratios

Capital adequacy requirements (Basel II & US):

- Tier 1 common ratio $\geq 2\%$.
- Tier 1 ratio $\geq 4\%$.
- Total capital ratio $\geq 8\%$.
- Tier 2 $\leq$ Tier 1.

Leverage ratio limits (US):

- Tangible assets = Assets - intangibles (like goodwill).
- Tier 1 / Tangible assets $\geq 3\%$.

Reason for moving liabilities off balance sheet.

The FRTB did away with risk weighted assets for the trading book, but subsequent Basel regulations brought it back!

So as a result, in this case, our *toy* bank is considered in compliance with the Capital Adequacy Requirements.

e)

As shown, the leverage ratio is defined as Tier 1 Capital over tangible assets, with a benchmark of 3% in the States.

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Reason for moving liabilities off balance sheet.

The FRTB did away with risk weighted assets for the trading book, but subsequent Basel regulations brought it back!

We have worked out in 2(b) that our Capital is 7,000, whereas our total tangible assets are 189,000. As a result:

$$LeverageRatio = \frac{7,000}{189,000} = 3.70\%$$

We note that in this case, the leverage ratio passes the US Leverage ratio limit.

Question 3:

3. Option portfolio VaR

A portfolio consists of a \$10,000,000 position in the S&P. Risk management says that the 99% 5 day VaR has to be reduced by 20%. Instead of liquidating 20% of the portfolio to do this, the portfolio manager would like to liquidate a portion of the portfolio and use the money to buy 1 year maturity ATM puts to bring his VaR down the required amount.

Use the data from the spreadsheet `hw10.csv`, which contains approximately the last 10 years of S&P prices and the corresponding implied volatilities for at the money 1 year puts. The spreadsheet was generated by `hw10.xls`, which downloaded the data from the Bloomberg. Assume the implied volatility surface is flat (i.e., that all options on the S&P on a given date trade at the same implied volatility, regardless of strike and maturity).

Assume the 1 year risk free rate is 0.5%.

Assume the only risk factor is the stock price (i.e. – ignore implied volatility risk).

- (a) What are the GBM parameters and the current VaR of the portfolio (on 11/4/2016), calibrating to a 5 year window?
 - (b) What is the VaR of the portfolio if 1% is liquidated and used to buy at the money 1 year maturity puts?
 - (c) What percentage of the portfolio should be liquidated to purchase ATM 1 year puts so as to reduce the portfolio VaR by 20%?
 - (d) How would the results change if volatility risk were taken into account?
-

a)

Given GBM, where we have the following:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t,$$

And that has the implication that over a small interval of Δt , we have the following:

$$\log\left(\frac{S_{t+\Delta t}}{S_t}\right) \sim N\left(\mu - \frac{1}{2}\sigma^2, \sigma^2\Delta t\right)$$

Then from, here we let have the daily log returns be r_i , and then we have the following:

$$\hat{\sigma}^2 = \frac{\text{Var}(r_i)}{\Delta t}, \quad \hat{\mu} = \frac{\mathbb{E}[r_i]}{\Delta t} + \frac{1}{2}\hat{\sigma}^2$$

with $\Delta t = 1/252$.

Then, we have following estimates from our modelled S&P:

	Item	Value
0	Estimated mu (annualized)	0.111012
1	Estimated sigma (annualized)	0.134344
2	5-day horizon h	0.019841
3	1% standard normal quantile $z_{0.01}$	-2.326348
4	1% stock-price quantile after 5 days	1,999.417788
5	Index units held	4,795.749048
6	Portfolio value at 1% tail stock level	9,588,705.951645
7	99% 5-day VaR	411,294.048355

As a result, we know that the current index level is:

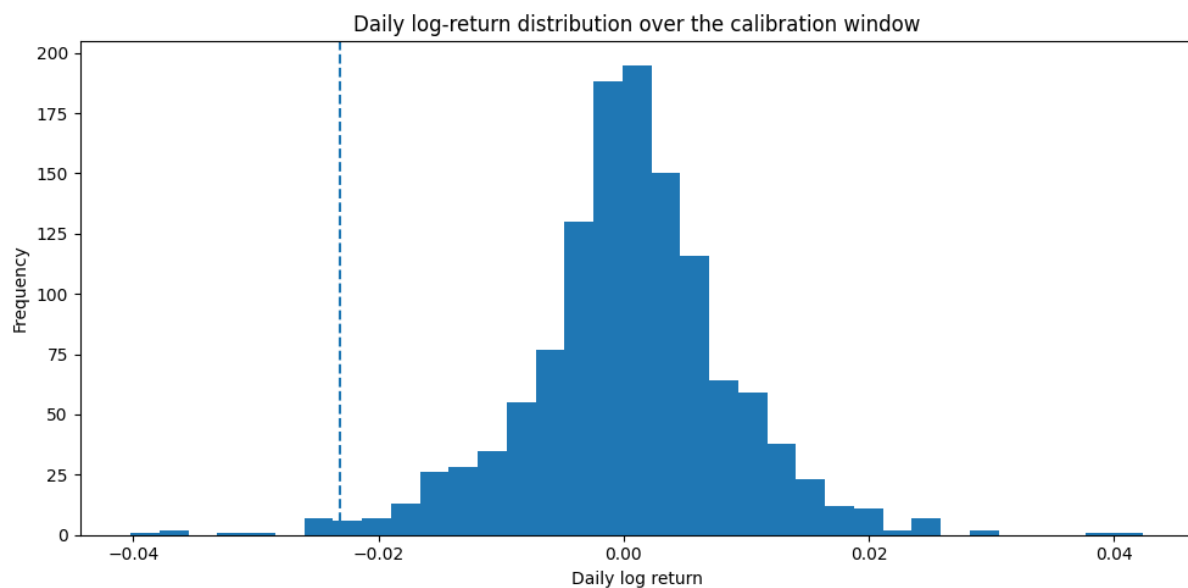
$$S_0 = 2085.18,$$

and the portfolio is a \$10,000,000 long position, so the number of index units held is

$$N_0 = \frac{10,000,000}{2085.18} = 4795.75$$

Then, we know that of for $z_{0.01} = -2.3263$. We know then from this, the 1 percent stock-price quantile is therefore

$$S_{0.01} = S_0 \exp\left(\left(\hat{\mu} - \frac{1}{2}\hat{\sigma}^2\right)h + \hat{\sigma}\sqrt{h} z_{0.01}\right) = 1999.42$$



That means the 1 percent tail portfolio value is

$$V_{0.01} = N_0 S_{0.01} = 9,588,705.95,$$

so the current 99 percent 5-day VaR is

$$\text{VaR}_{0.99} = 10,000,000 - 9,588,705.95 = \$ 411,294.$$

As a result, in this pure long portfolio, the VaR will entirely come from the left-tail move in the index. As such over 5 trading days, the vol-term will matter much more than the drift estimate.

b)

From here, it is stated that 1% of the stock portfolio is liquidated to purchase 1y ATM puts, which means that:

$$\alpha = 0.01, \alpha V_0 = 100,000$$

And then, we know that:

$$(1 - \alpha)V_0 = 9,900,000$$

We know to avoid the implied volatility, and assume a flat IV surface. So now, the put is repriced under the tail stock scenario, with the same IV with stock and time to maturity changed.

At the date of 04/11/2016, we have:

$$\sigma_{imp} = 17.67$$

And the risk-free rate is $r = 0.5\%$, and since it is ATM at inception, we know:

$$K = S_0 = 2085.18$$

Therefore, from here, we know the BSM put formula is:

$$P(S, K, r, \sigma, T) = Ke^{-rT}N(-d_2) - SN(-d_1),$$

With the following:

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T},$$

the initial put premium (market price) is

$$P_0 = 141.292245.$$

So the number of puts purchased is

$$N_P = \frac{100,000}{141.292245} = 707.7529.$$

Rounding down. Notably the remaining stock unites as a result will be the following:

$$N_S = \frac{9,900,000}{2085.18} = 4747.791558.$$

Now move to the same 1 percent tail stock scenario from part (a), so $S = S_{0.01} = 1999.417788$, and reduce maturity by 5 trading days of the following:

$$T_{\text{rem}} = 1 - \frac{5}{252}$$

Then, if we keep the IV fixed, the put is worth

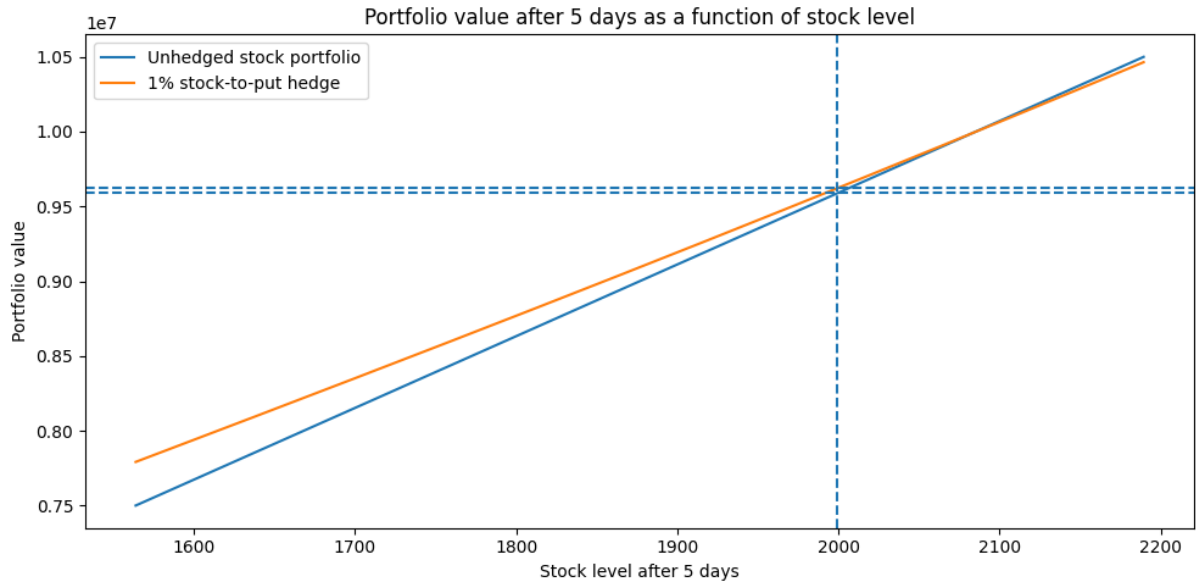
$$P_{0.01} = 182.952279$$

in that downside scenario. Therefore, the tail value of the hedged portfolio is

$$V_{0.01}^{\text{hedged}} = N_S S_{0.01} + N_P P_{0.01} = 9,622,303.90$$

So the hedged VaR is

$$\text{VaR}_{0.99}^{\text{hedged}} = 10,000,000 - 9,622,303.90 = \$377,696.10$$



This is lower than the unhedged VaR because the put appreciates in the downside scenario. The reduction in VaR is

$$411,294.05 - 377,696.10 = 33,597.95,$$

And when we consider it in % terms, we get the following:

$$\frac{33,597.95}{411,294.05} = 8.17\% \text{ (2 d.p.)}$$

So liquidating 1 percent into puts helps, but not enough to reach the required 20 percent VaR reduction. So, we can consider potentially some optimisation.

c)

Here, we can let α denote the fraction liquidated from stock and reallocated into puts. Then the stock units held become

$$N_S(\alpha) = \frac{(1 - \alpha)V_0}{S_0},$$

the number of puts purchased becomes

$$N_P(\alpha) = \frac{\alpha V_0}{P_0},$$

and the tail portfolio value is

$$V_{0.01}(\alpha) = N_S(\alpha)S_{0.01} + N_P(\alpha)P_{0.01}.$$

Hence the hedged VaR as a function of α is

$$\text{VaR}(\alpha) = V_0 - V_{0.01}(\alpha).$$

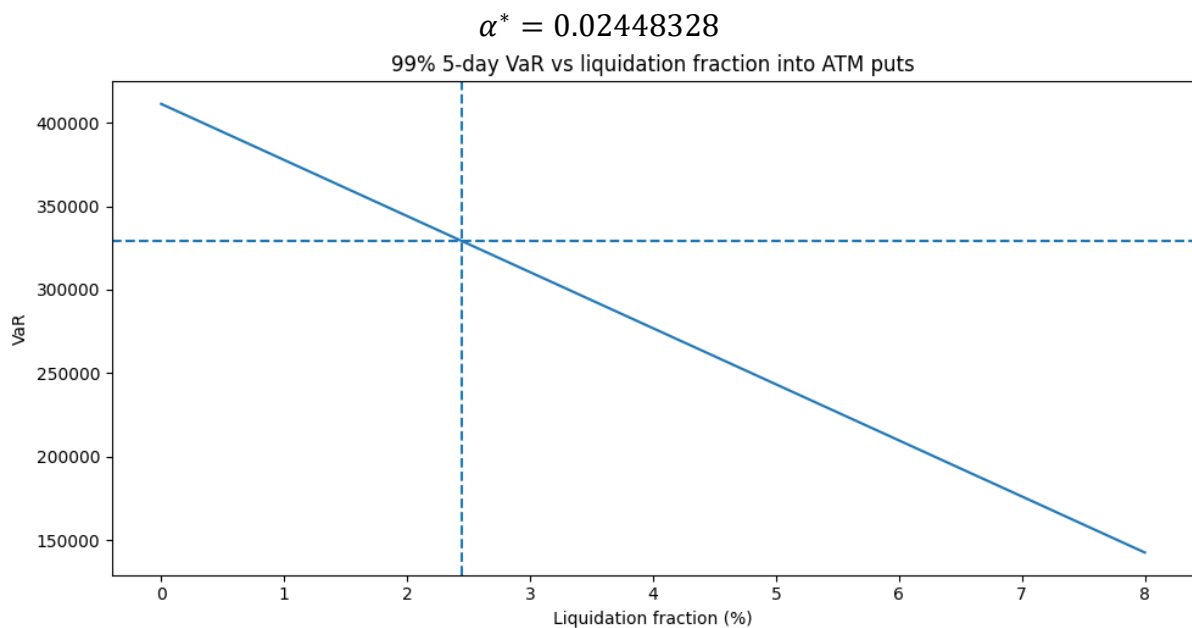
The target is a 20 percent reduction in the original VaR, so the required target level is

$$0.8 \times 411,294.05 = 329,035.24.$$

The notebook solves

$$\text{VaR}(\alpha) = 329,035.24$$

numerically by bisection (root-finding) and finds



This becomes an optimisation question, and bisection was used, although Newton, or Brent from SciPy could have been used.

So the required liquidation percentage is about 2.45%, rounded of to 2d.p. We can conclude from this that the 1% hedge is insufficient, and we need to put about 2.45% of net assets in to ATM 1y puts to cut the 99% VaR by 20%.

d)

From the previous parts, the put is repriced in a downside scenario with stock and time changed, the the IV remains fixed, so the tail put value becomes:

$$P_{0.01} = P(S_{0.01}, K, r, \sigma_{\text{imp}}, T_{\text{rem}}).$$

If volatility risk is allowed, then instead the put value is

$$P_{0.01} = P(S_{0.01}, K, r, \sigma_{0.01}, T_{\text{rem}}),$$

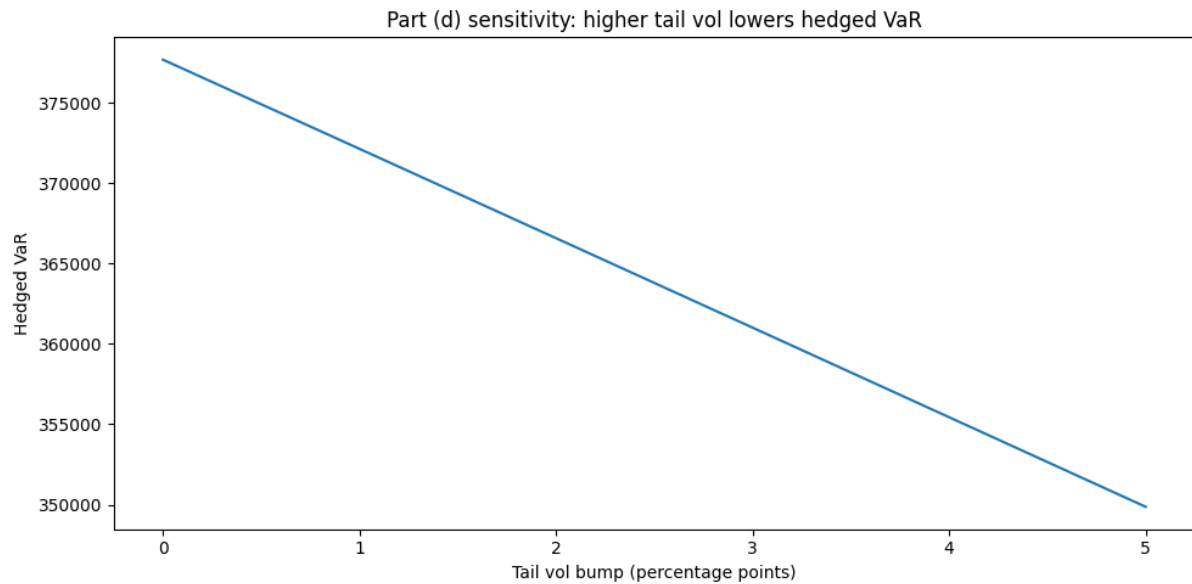
where $\sigma_{0.01}$ can move with the market. So the risk will now materially differ, since we may either be under or over-hedged, and the error term is driven by vega. This is because the put value now depend on the second state variable σ , in addition to S . Gamma is still important, for the stock-direction nonlinearity of the put, but the incremental error from allowing IV to move is mainly a volatility-sensitivity effect rather than a pure gamma effect.

This means that the VaR calculation is changed in a material fashion, since the option hedge is now exposed to both the downside stock move and the associated vol move. Since we know that a long put has positive vega, an increase in IV will raise the put value and improve the hedge, whenever the stock portfolio is losing money. In equity markets, downside stock moves tend to be associated with higher IV, so allowing σ to rise in the left tail, will naturally make the hedge more effective than fixing the volatility. The Fixed IV calculation, as a result, is likely more conservative w.r.t to the protective value of the put in stressed downside scenarios.

	Tail implied vol	Vol bump	Tail put value	Hedged VaR with 1% liquidation
0	0.176700	0.000000	182.952279	377,696.096940
1	0.186700	0.010000	190.795927	372,144.731984
2	0.196700	0.020000	198.652970	366,583.886589
3	0.206700	0.030000	206.520767	361,015.430046
4	0.226700	0.050000	222.280191	349,861.652094

We built a sensitivity table, where we note that with no vol bump, the tail implied volatility remains 0.1767, the tail put value is 182.95, and the hedged VaR with 1 percent liquidation is \$377,696.10.

Wherein, if the tail implied probability volatility is bumped upward by 1 vol point (1 bp) the tail put value rises to 190.80, and the hedged VaR falls to about \$372,144.74. For a 5 vol bump, the VaR falls to \$349K-ish. Hence the directional idea showcases that a higher tail implied volatility increases the value of the protective put and lowers the portfolio's hedged VaR.



Noting it in this diagram above, it shows that it scales almost linearly over the small bump range, although, in a fuller model, the relationship is probably not exactly linear, as under BSM, option values are non-linear in volatility, and neither would it be in other models like Heston.

So resultingly, we know that once volatility risk is incorporated, the hedge gains an additional source of protection via Vega. As a result, the VaR computed in previous parts will generally be overstated relative to a more realistic joint stock volatility model. Conclusively if downside equity moves, are accompanied by a vol spike (which it does, usually – consider VIX during Covid cash), then fewer puts will be needed to achieve the same target VaR reduction.

HW10 Q3

```
In [4]: import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from statistics import NormalDist

pd.set_option("display.float_format", lambda x: f"{x:,.6f}")

# Load and clean data
csv_path = "data/hw10.csv"
raw = pd.read_csv(csv_path)

df = raw.rename(columns={
    raw.columns[0]: "Date",
    raw.columns[1]: "PX_LAST",
    raw.columns[2]: "IMP_VOL"
}).iloc[1:].copy()

df["Date"] = pd.to_datetime(df["Date"])
df["PX_LAST"] = pd.to_numeric(df["PX_LAST"])
df["IMP_VOL"] = pd.to_numeric(df["IMP_VOL"]) / 100.0

# Inputs
end_date = pd.Timestamp("2016-11-04")
start_date = end_date - pd.DateOffset(years=5)
V0 = 10_000_000
r = 0.005
T = 1.0
dt = 1 / 252
h = 5 / 252
alpha_1pct = 0.01
z01 = NormalDist().inv_cdf(0.01)

# Calibration window
window = df[(df["Date"] >= start_date) & (df["Date"] <= end_date)].copy()
window["logret"] = np.log(window["PX_LAST"] / window["PX_LAST"].shift(1))
rets = window["logret"].dropna()

mu_hat = rets.mean() / dt + 0.5 * rets.var(ddof=0) / dt
sigma_hat = math.sqrt(rets.var(ddof=0) / dt)

S0 = float(df.loc[df["Date"] == end_date, "PX_LAST"].iloc[0])
iv0 = float(df.loc[df["Date"] == end_date, "IMP_VOL"].iloc[0])
K = S0
N0 = V0 / S0

def bs_put(S, K, r, sigma, T):
    N = NormalDist()
    d1 = (math.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    return K * math.exp(-r * T) * N.cdf(-d2) - S * N.cdf(-d1)

S_q = S0 * math.exp((mu_hat - 0.5 * sigma_hat**2) * h + sigma_hat * math.sqrt(h))
V_q_unhedged = N0 * S_q
```

```

var_unhedged = V0 - V_q_unhedged

P0 = bs_put(S0, K, r, iv0, T)
T_rem = T - h
P_q_constvol = bs_put(S_q, K, r, iv0, T_rem)

N_stock_1pct = (1 - alpha_1pct) * V0 / S0
N_put_1pct = alpha_1pct * V0 / P0
V_q_hedged_1pct = N_stock_1pct * S_q + N_put_1pct * P_q_constvol
var_hedged_1pct = V0 - V_q_hedged_1pct

def hedged_var(alpha, sigma_tail=iv0):
    N_stock = (1 - alpha) * V0 / S0
    N_put = alpha * V0 / P0
    P_tail = bs_put(S_q, K, r, sigma_tail, T_rem)
    V_tail = N_stock * S_q + N_put * P_tail
    return V0 - V_tail

target_var = 0.8 * var_unhedged

lo, hi = 0.0, 0.20
for _ in range(200):
    mid = 0.5 * (lo + hi)
    if hedged_var(mid) > target_var:
        lo = mid
    else:
        hi = mid
alpha_star = 0.5 * (lo + hi)
var_at_alpha_star = hedged_var(alpha_star)

print("Loaded and calibrated.")

```

Loaded and calibrated.

```

In [5]: summary = pd.DataFrame({
    "Quantity": [
        "Calibration window start",
        "Calibration window end",
        "Observations in 5-year window",
        "Current SPX level S0",
        "Current ATM 1Y implied vol",
        "Annualized mu hat",
        "Annualized sigma hat",
        "Portfolio value V0",
        "5-day horizon h",
        "1% normal quantile z0.01"
    ],
    "Value": [
        start_date.date(),
        end_date.date(),
        len(rets),
        S0,
        iv0,
        mu_hat,
        sigma_hat,
        V0,
        h,
        z01
    ]
})

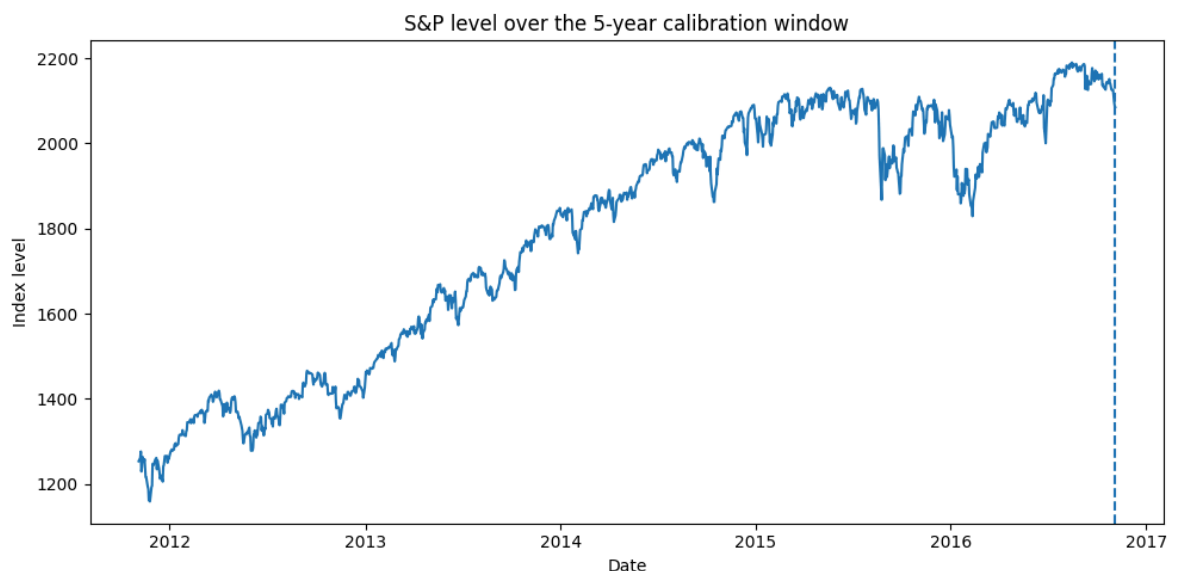
```

```
})
summary
```

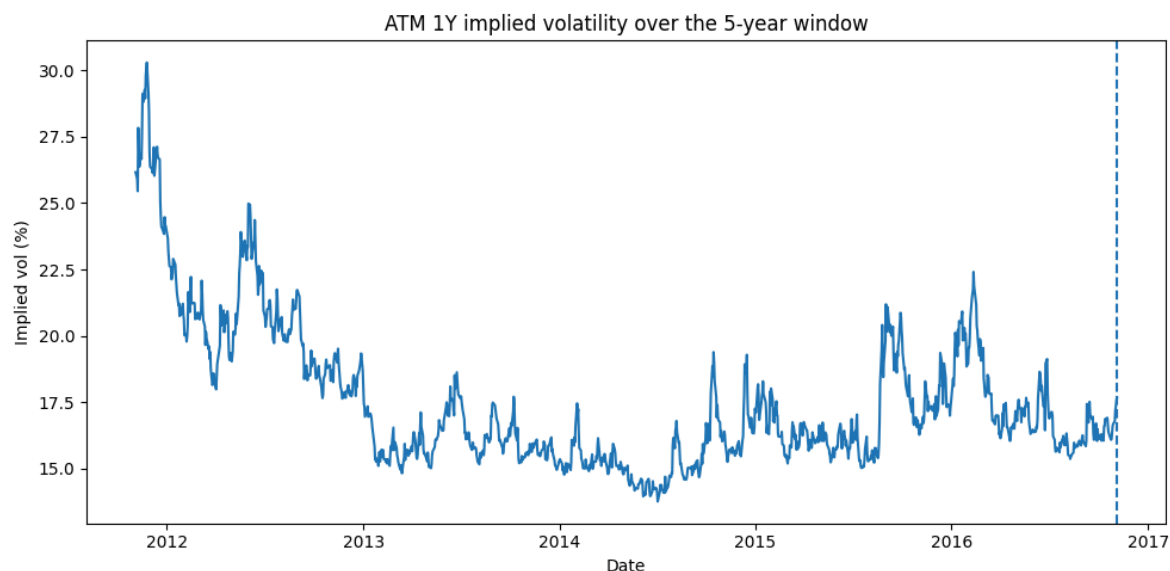
Out [5]:

	Quantity	Value
0	Calibration window start	2011-11-04
1	Calibration window end	2016-11-04
2	Observations in 5-year window	1258
3	Current SPX level S0	2,085.180000
4	Current ATM 1Y implied vol	0.176700
5	Annualized mu hat	0.111012
6	Annualized sigma hat	0.134344
7	Portfolio value V0	10000000
8	5-day horizon h	0.019841
9	1% normal quantile z0.01	-2.326348

```
In [6]: plt.figure(figsize=(10, 5))
plt.plot(window["Date"], window["PX_LAST"])
plt.axvline(end_date, linestyle="--")
plt.title("S&P level over the 5-year calibration window")
plt.xlabel("Date")
plt.ylabel("Index level")
plt.tight_layout()
plt.show()
```



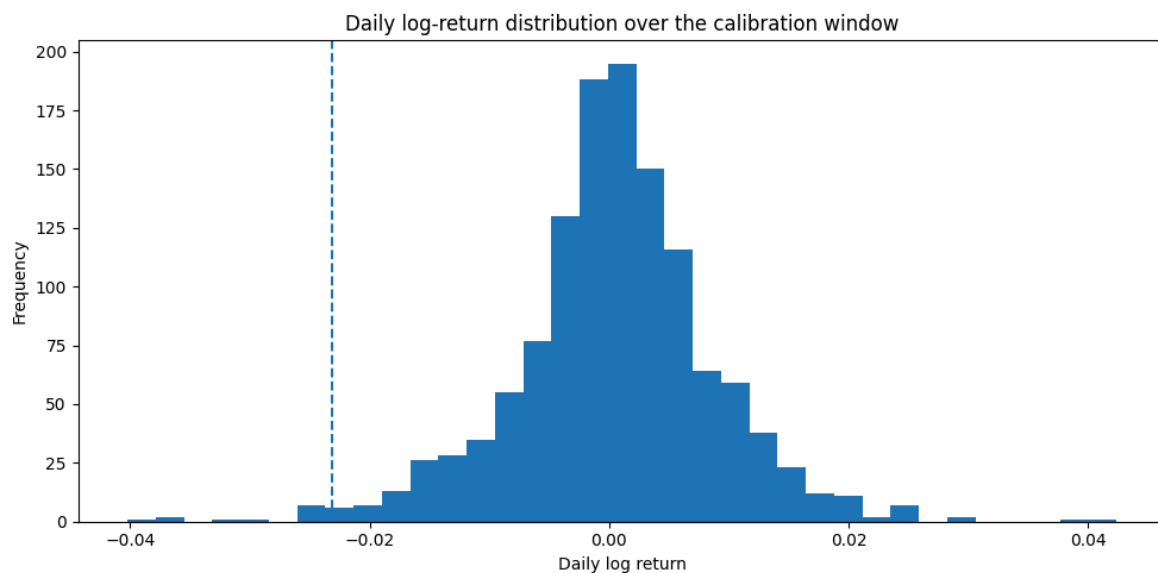
```
In [7]: plt.figure(figsize=(10, 5))
plt.plot(window["Date"], 100 * window["IMP_VOL"])
plt.axvline(end_date, linestyle="--")
plt.title("ATM 1Y implied volatility over the 5-year window")
plt.xlabel("Date")
plt.ylabel("Implied vol (%)")
plt.tight_layout()
plt.show()
```



```
In [8]: q01_ret = np.quantile(rets, 0.01)

plt.figure(figsize=(10, 5))
plt.hist(rets, bins=35)
plt.axvline(q01_ret, linestyle="--")
plt.title("Daily log-return distribution over the calibration window")
plt.xlabel("Daily log return")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

part_a = pd.DataFrame({
    "Item": [
        "Index units held  $N_0 = V_0 / S_0$ ",
        "1% tail stock price after 5 days",
        "Tail portfolio value (unhedged)",
        "99% 5-day VaR"
    ],
    "Value": [
        N0,
        S_q,
        V_q_unhedged,
        var_unhedged
    ]
})
part_a
```



Out [8]:

	Item	Value
0	Index units held $N_0 = V_0 / S_0$	4,795.749048
1	1% tail stock price after 5 days	1,999.417788
2	Tail portfolio value (unhedged)	9,588,705.951645
3	99% 5-day VaR	411,294.048355

```
In [9]: part_b = pd.DataFrame({
    "Item": [
        "Fraction liquidated",
        "Dollars moved into puts",
        "Initial ATM put price  $P_0$ ",
        "Stock units remaining",
        "Put units purchased",
        "Put value at tail stock level",
        "Tail portfolio value with 1% hedge",
        "99% 5-day VaR with 1% hedge",
        "Dollar VaR reduction",
        "Percentage VaR reduction"
    ],
    "Value": [
        alpha_1pct,
        alpha_1pct * V0,
        P0,
        N_stock_1pct,
        N_put_1pct,
        P_q_constvol,
        V_q_hedged_1pct,
        var_hedged_1pct,
        var_unhedged - var_hedged_1pct,
        (var_unhedged - var_hedged_1pct) / var_unhedged
    ]
})
part_b
```

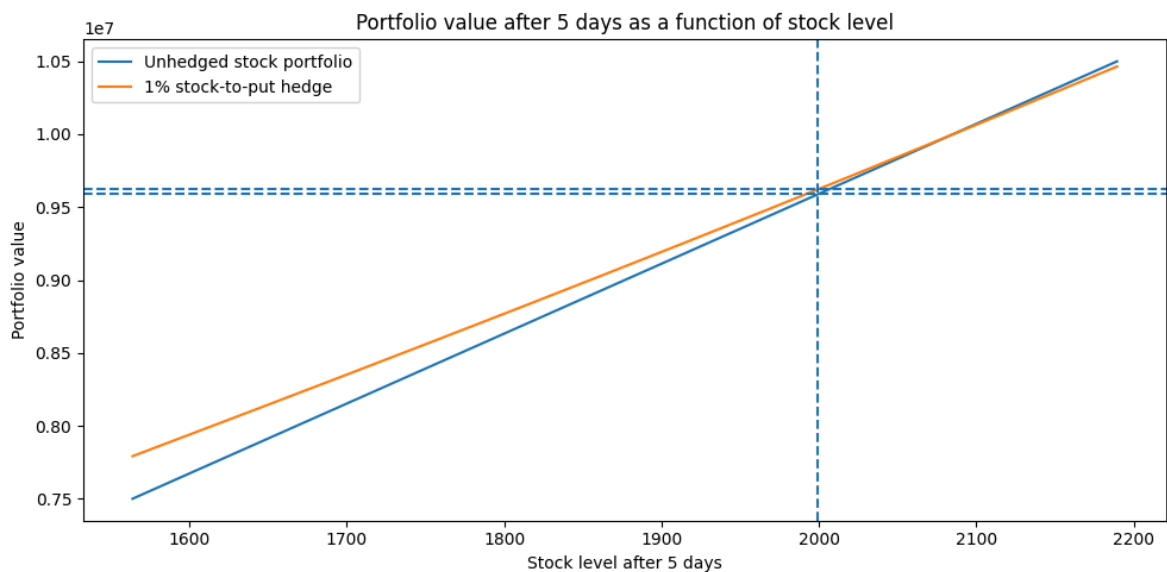
Out [9]:

	Item	Value
0	Fraction liquidated	0.010000
1	Dollars moved into puts	100,000.000000
2	Initial ATM put price P0	141.292245
3	Stock units remaining	4,747.791558
4	Put units purchased	707.752928
5	Put value at tail stock level	182.952279
6	Tail portfolio value with 1% hedge	9,622,303.903060
7	99% 5-day VaR with 1% hedge	377,696.096940
8	Dollar VaR reduction	33,597.951415
9	Percentage VaR reduction	0.081688

```
In [10]: stock_grid = np.linspace(0.75 * S0, 1.05 * S0, 300)
put_grid = np.array([bs_put(s, K, r, iv0, T_rem) for s in stock_grid])

unhedged_grid = N0 * stock_grid
hedged_grid = N_stock_1pct * stock_grid + N_put_1pct * put_grid

plt.figure(figsize=(10, 5))
plt.plot(stock_grid, unhedged_grid, label="Unhedged stock portfolio")
plt.plot(stock_grid, hedged_grid, label="1% stock-to-put hedge")
plt.axvline(S_q, linestyle="--")
plt.axhline(V_q_unhedged, linestyle="--")
plt.axhline(V_q_hedged_1pct, linestyle="--")
plt.title("Portfolio value after 5 days as a function of stock level")
plt.xlabel("Stock level after 5 days")
plt.ylabel("Portfolio value")
plt.legend()
plt.tight_layout()
plt.show()
```



```
In [11]: part_c = pd.DataFrame({
    "Item": [
```

```

        "Unhedged VaR",
        "Target VaR = 80% of unhedged VaR",
        "Required liquidation fraction alpha*",
        "Required liquidation percentage",
        "VaR at alpha*"
    ],
    "Value": [
        var_unhedged,
        target_var,
        alpha_star,
        100 * alpha_star,
        var_at_alpha_star
    ]
})
part_c

```

Out [11]:

	Item	Value
0	Unhedged VaR	411,294.048355
1	Target VaR = 80% of unhedged VaR	329,035.238684
2	Required liquidation fraction alpha*	0.024483
3	Required liquidation percentage	2.448328
4	VaR at alpha*	329,035.238684

```

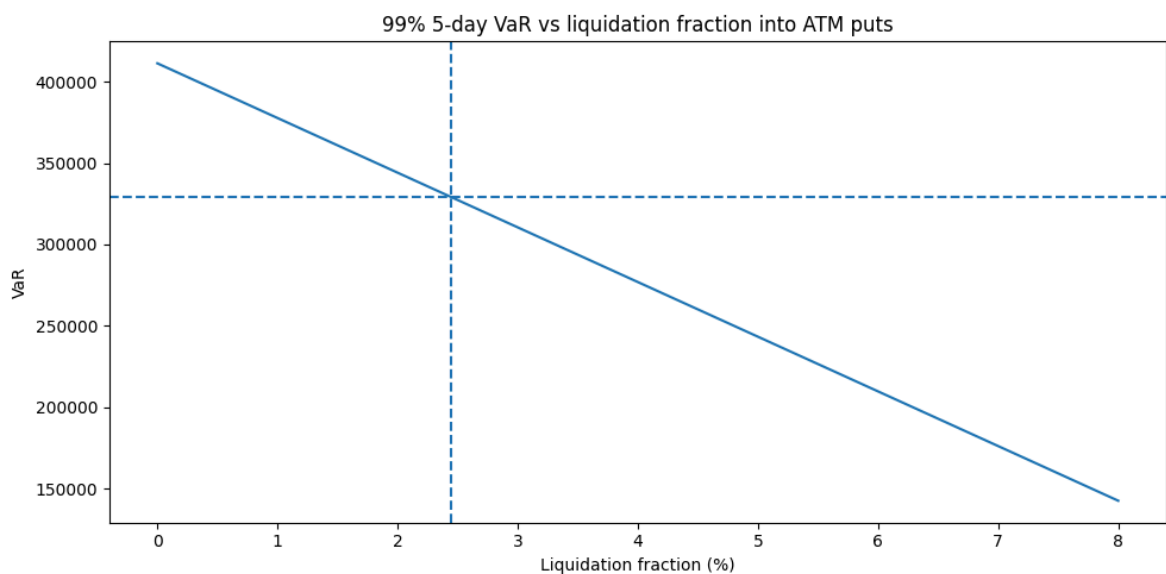
In [12]: alphas = np.linspace(0, 0.08, 200)
var_curve = np.array([hedged_var(a) for a in alphas])

```

```

plt.figure(figsize=(10, 5))
plt.plot(100 * alphas, var_curve)
plt.axhline(target_var, linestyle="--")
plt.axvline(100 * alpha_star, linestyle="--")
plt.title("99% 5-day VaR vs liquidation fraction into ATM puts")
plt.xlabel("Liquidation fraction (%)")
plt.ylabel("VaR")
plt.tight_layout()
plt.show()

```



```

In [13]: vol_sensitivity = []
for bump in [0.00, 0.01, 0.02, 0.03, 0.05]:
    sigma_tail = iv0 + bump
    P_tail = bs_put(S_q, K, r, sigma_tail, T_rem)
    var_tail = hedged_var(0.01, sigma_tail=sigma_tail)
    vol_sensitivity.append({
        "Tail implied vol": sigma_tail,
        "Vol bump": bump,
        "Tail put value": P_tail,
        "Hedged VaR with 1% liquidation": var_tail
    })

vol_sensitivity_df = pd.DataFrame(vol_sensitivity)
vol_sensitivity_df

```

```

Out[13]:
   Tail implied vol  Vol bump  Tail put value  Hedged VaR with 1% liquidation
0          0.176700  0.000000      182.952279             377,696.096940
1          0.186700  0.010000      190.795927             372,144.731984
2          0.196700  0.020000      198.652970             366,583.886589
3          0.206700  0.030000      206.520767             361,015.430046
4          0.226700  0.050000      222.280191             349,861.652094

```

```

In [15]: plt.figure(figsize=(10, 5))
plt.plot(100 * vol_sensitivity_df["Vol bump"], vol_sensitivity_df["Hedged VaR"])
plt.title("Part (d) sensitivity: higher tail vol lowers hedged VaR")
plt.xlabel("Tail vol bump (percentage points)")
plt.ylabel("Hedged VaR")
plt.tight_layout()
plt.show()

```

